



ROBOTECH VD PRIVATE LIMITED

INTERNSHIP FINAL TEAM PROJECT

AI-Based Autonomous Robot Monitoring & Navigation System

1. Project Overview

This multidisciplinary internship capstone brings the interns together as one team to design and demonstrate a small autonomous mobile-robot system. It combines robotics, AI/ML, software development, mobile application development, and technical research. It is a separate internship project and is not intended to represent or disclose any ongoing ROBOTECH company project.

Target outcome: a working prototype or controlled demonstration in which a mobile robot can sense obstacles, receive/control movement, perform a defined AI perception task, and expose its status through a software/mobile interface.

2. Problem Statement

The project addresses the challenge of integrating sensing, decision-making, communication, and human-facing control into one small robotic system. The team will build one integrated prototype rather than six unrelated individual projects.

3. Core Objectives

- Build or simulate a small mobile robotic platform with basic movement.
- Implement obstacle sensing and a simple navigation/decision layer.
- Develop a limited AI perception function, such as recognition of selected objects.
- Create a software communication layer for exchanging robot status and commands.
- Create a mobile interface for basic monitoring and control.
- Document architecture, methodology, experiments, results, limitations, and future improvements.

4. System Architecture

Layer	Main function
Mobile App	Robot status, basic control, alerts/results
Backend / Communication	API, data exchange, command handling
Robot Controller	Motor commands, sensor reading, control logic
Sensors	Obstacle/distance information
Camera + AI/ML	Selected object/perception task
Mechanical Platform	Chassis, motors, wheels and power

Data flow: Sensors/Camera → Robot Controller → AI/Decision Layer → Communication Layer → Mobile App; commands travel in the reverse direction.

5. Team Responsibilities

Intern	Offer-letter role	Capstone responsibility
Likhitha	AI/ML Intern	Object/perception model; dataset preparation; model testing and metrics.
Akanksha	AI/ML Intern	AI pipeline; preprocessing; model comparison and integration support.
Sathwika	AI/ML Intern	Real-time inference/testing; performance analysis; AI integration support.
Sasamrutha	Research & AI Intern	Literature review; methodology; experiment design; technical documentation.
Poojitha	Software Development Intern	Backend/API; communication layer; command/data handling; integration.
Svitha	Mobile Application Development Intern	Mobile dashboard; robot controls; status/result display.

6. Suggested Technology Stack

Component	Suggested options
Controller	ESP32 / Raspberry Pi, depending on available hardware
Sensors	Ultrasonic or ToF distance sensor; optional IMU
Camera	USB / Raspberry Pi camera / compatible module
AI/ML	Python, OpenCV, TensorFlow Lite or PyTorch; choose one
Backend	Python FastAPI/Flask or Node.js
Mobile	Flutter or Android Studio
Communication	Wi-Fi; REST API or WebSocket
Version control	Git + GitHub

7. Minimum Viable Demonstration

- Robot moves through basic manual commands.
- Distance sensor detects an obstacle and the system responds with a defined action.
- AI model recognizes at least 2–3 selected classes/objects under controlled conditions.
- Backend receives robot data and sends commands.
- Mobile app displays robot status and provides basic control.
- The team demonstrates the complete data flow from robot to app and back.

8. Project Schedule: 22 August – 5 September

Period	Focus	Expected output
22–23 Aug	Kick-off, requirements, architecture	Final scope, responsibilities, Git repository
24–26 Aug	AI + sensor + controller modules	Initial AI model and sensor/robot control tests
27–29 Aug	Backend + communication	Working API/communication path
30 Aug–1 Sep	Mobile application	Basic dashboard and control interface
2–3 Sep	Integration + testing	End-to-end prototype/demo
4 Sep	Documentation + presentation	Report, PPT, diagrams, results
5 Sep	Final evaluation	Final demo, viva and submission

9. Individual Deliverables

- AI/ML interns: dataset, model notebook/source, evaluation metrics, inference demo and integration notes.
- Research & AI: literature review, methodology, comparison, testing protocol, limitations and references.
- Software Development: source code, API documentation, communication protocol and integration notes.
- Mobile Development: application source, UI screenshots, control/status implementation and build/demo.
- All interns: individual contribution summary, weekly progress record and final presentation section.

10. Final Submission Package

- Complete project source code organized by module.
- Git repository with meaningful commit history.
- System architecture and data-flow diagrams.
- Final project report.
- Final presentation (PPT).
- Working demonstration/video if hardware access is limited.
- Individual contribution report for every intern.
- Testing results, screenshots and limitations.

11. Evaluation Criteria

Area	Weight
Technical implementation	25%
Individual contribution	20%
Integration and teamwork	15%
Testing and results	15%
Documentation	10%
Presentation and viva	15%

12. Scope Control

Important: This is an internship capstone, not a commercial product. Prioritize a small, working end-to-end demonstration over a large feature list. Safety-critical autonomy, medical use, surveillance, and production deployment are outside this project scope.

Final project statement: The team will demonstrate how AI/ML, software, mobile development, research, and robotics can be integrated into one controlled autonomous-robot prototype while preserving clear individual responsibilities.